

Combined hormonal contraception

(From FSRH Guideline, 2019)



Red Whale

GEMS

Guidelines & Evidence Made Simple

What is CHC (combined hormonal contraception)?

- Contains an oestrogen and a progestogen. These work synergistically to prevent ovulation, render cervical mucus hostile to sperm and keep endometrium thin and stable. Preparations differ in type/dose of oestrogen/progestogen.
- Includes the combined pill, patch and vaginal ring.
- **Oestrogen:** most CHC contains synthetic ethinyl oestradiol, but some contains mestranol, estetrol or oestradiol.
- **Progestogen:** classified according to 'generation' which reflects introduction onto the CHC market (see table below). Newer generation progestogens were developed to have fewer androgenic/glucocorticoid effects and so may be better tolerated BUT studies suggest they may incur a higher VTE risk.

Benefits of CHC

- Light, predictable bleeding patterns.
- Less dysmenorrhea.
- Manages symptoms of:
 - PMS.
 - PCOS.
 - Endometriosis (use continuous regime).
 - Menopause (can use up to 50y).
- Reduces risk of endometrial, ovarian, colorectal cancer.

Progestogens

1 st generation	Norethisterone
2 nd generation	Levonorgestrel
3 rd generation	Desogestrel Gestodene Etonogestrel Norgestimate (sometimes considered 2 nd generation as metabolised to levonorgestrel) Norelgestomin
4 th generation	Drospirenone Dienogest Nomogestrel

Efficacy of CHC

- **CHC is not as effective as long-acting methods: typical failure rate is 9% in first year of use!**
- Highly user-dependent! Remember to follow missed pill rules if vomits within 3h of taking/has diarrhoea >24h.
- No additional precautions needed if taking non-enzyme-inducing antibiotics unless severe vomiting or diarrhoea.

Contraindications and cautions

Common absolute contraindications (UKMEC 4)

- BP $\geq 160/100$.
- Current breast cancer.
- Smoking ≥ 15 cigarettes/d AND ≥ 35 y.
- Current vascular disease.
- History of VTE or known thrombogenic mutation (e.g. factor 5 Leiden, prothrombin mutation, protein S or C, antithrombin deficiencies).
- Migraine with aura.

Always check www.fsrh.org/ukmec if unsure whether a medical condition or lifestyle factor restricts CHC use.

Cautions to use

- BMI ≥ 35 is UKMEC 3 (relative contraindication) (CHC efficacy unaffected by weight, except for patch if >90 kg).
- Patients on teratogens should be on a LARC (as CHC not effective enough).
- Avoid CHC if taking enzyme inducers or lamotrigine (can lower seizure threshold).
- Avoid pills in malabsorptive state (e.g. post-bariatric surgery).

Risks (small but greater than with other methods)

Counselling about VTE (from FSRH guideline)

- Risk increases around 3x on CHC (but much lower than in pregnancy).
- 1st/2nd generation formulations appear lower risk than 3rd/4th generation.
- Absolute risk is very small. Annual risk of VTE on CHC $\approx 5-12/10\ 000$ compared with $2/10\ 000$ in non-CHC users.
- Approximately 1% are fatal.

Other (small) risks

- Arterial thrombosis (higher risk with higher oestrogen doses).
- Breast and cervical cancer (risk reduces after stopping).

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Starting combined hormonal contraception

(From FSRH Guideline, 2019)

Regimes

Traditional (licensed) regimens: involve 21d of pill-taking then a 7d break.

Tailored or extended regimens: recommended by FSRH as involve fewer, shorter or no hormone-free intervals, and therefore potentially less bleeding, fewer side-effects and lower risk of contraceptive failure. Note: the patch and ring may be used in the same way, but phasic pill preparations should NOT.

- **Scheduled extended cycles (bi/tricycling):** take 2–3 packs consecutively followed by a 4–7d hormone-free interval.
- **Unscheduled extended cycle:** take pills continuously for >21d, after which, when/if breakthrough bleeding occurs, take a 4–7d break.
- **Continuous pill-taking:** take a pill daily whether bleeding occurs or not.

Before starting CHC

Before prescribing:

- Check medical eligibility (medical history, lifestyle factors, drug and family history).
- Measure and record BP and BMI.
- Counsel regarding efficacy, risks, benefits, side-effects and regimens...

Which formulation and how much?

- Prescribe $\leq 30\text{mcg}$ pill with a 1st or 2nd generation progestogen first line.
- Consider an alternative formulation if likely to be more suitable based on patient experience, tolerance or preference.
- Give up to 12m supply (except vaginal ring: prescribe only 3m due to shelf life).

Starting instructions

If no risk of pregnancy, start CHC immediately:

Up to day 5 of natural menstrual cycle: can be started without using extra precautions.

- **After day 5 of menstrual cycle:** use extra precautions for 7d.
- **If swapping from alternative form of contraception:** use extra precautions for 7d if the previous method was not anti-ovulant (implant, depot, POP).

After oral emergency contraception:

- **Levonorgestrel emergency contraception:** start immediately after giving levonorgestrel, use extra precautions for 7d and do a pregnancy test at 3w.
- **Ulipristal emergency contraception:** **do NOT start CHC until 5d after giving ulipristal**, then use extra precautions for a further 7d and do a pregnancy test at 3w.
 - EXCEPTION: if an established CHC user misses 2–7d of pills/patch/ring in the first week after the hormone-free interval, we can give ulipristal as emergency contraception and she can restart CHC immediately. She should use extra precautions for 7 days and do a pregnancy test at 3w.

Review annually

- Check for compliance and any problems.
- Reassess eligibility (including BP, BMI and smoking status).
- Continue up to 50y if acceptable.
- Stopping and starting the pill every few months/years to have a 'pill holiday' is not recommended and may actually increase overall risk of VTE.

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Troubleshooting tips

Breakthrough bleeding

- Common! Usually settles in 3–4m.
- Higher oestrogen preparations may be associated with less bleeding.
- 2nd generation formulations may be better than 1st generation.
- Extended or continuous regimens may be better.
- Consider other causes (e.g. STIs, pregnancy, compliance, cervical/uterine pathology).

Headache

- No optimal preparation recommended.
- Extended/continuous regimes may be better if occurs in hormone-free interval.

Mood changes

- Counsel that may be multifactorial.
- There is no clear evidence that CHC causes depression.
- Consider switching to a formulation with a different progestogen.
- Extended/continuous regimes may be better if symptoms occur in hormone-free interval.

Weight gain

- There is no known causal association between CHC and weight gain.
- There is no evidence that different formulations impact weight differently.

Low libido

- Evidence contradictory; overall, no clear effect of CHC on libido. Try different formulation or method of contraception.

Missed pill rules...

Has the hormone-free interval been extended >7d?

Yes

If UPSI has occurred in the hormone-free interval (HFI):
consider emergency contraception

No

≥2 missed pills (≥72h since last pill)

1 missed pill (48–<72h since last pill)

- **Emergency contraception NOT needed**
- Take missed pill ASAP
- Continue rest of pack
- NO additional precautions required

Week 1 after hormone-free interval (HFI)
Consider emergency contraception if UPSI has occurred in past week or in HFI

Week 2 or 3 after hormone-free interval (HFI)
(also applies for subsequent consecutive weeks of continuous pill-taking)
Emergency contraception NOT required

- Take the missing pill ASAP
- Continue rest of pack
- Use condoms/avoid sex for 7d
- Consider follow-up pregnancy test

- Take the missing pill ASAP
- Continue to take remaining pills in pack
- **If ≥2 pills missed in the 7d prior to a scheduled HFI: omit HFI**
- Use condoms/avoid sex until 7d of consecutive pill-taking

Combined hormonal contraception: formulations

(From MIMS 2022)



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20mcg ethinyl oestradiol monophasic pills

Progestogen	Name	Cost
Third generation		
Desogestrel 150mg	Bimizza	££
	Gedarel 20	££
	Mercilon	££
Gestodene 75mcg	Femodette	££
	Millinette 20	££
	Sunya	££
Fourth generation		
Drospirenone 3mg	Eloine	£££

Approximate prices per month:

£ = usually <£5

££ = usually £5–10

£££ = usually £10–15

££££ = usually £15–20

£££££ = usually >£20

30mcg ethinyl oestradiol monophasic pills

Progestogen	Name	Cost
Second generation		
Levonorgestrel 150mcg	Ambelina	£
	Elevin	££££££
	Levest	£
	Maexeni	£
	Microgynon (ED)	£
	Ovranette	£
	Rigevidon	£
Third generation		
Desogestrel 150mcg	Cimizt	£
	Gedarel 30	£
	Marvelon	££
Gestodene 75mcg	Femodene	££
	Katya	££
	Millinette 30	£
Fourth generation		
Drospirenone 3mg	Dretine	££
	Lucette	£££
	Yacella	££
	Yasmin	£££

35mcg equivalent monophasic pills

Oestrogen	Progestogen	Name	Cost
35mcg ethinyl oestradiol	First generation		
	Norethisterone 500mcg	Brevinor	£
	Norethisterone 1mg	Norimin	£
	Third generation		
	Norgestimate 250mcg	Cilique Lizinna	£ £
Mestranol 50mcg*	First generation		
	Norethisterone 1mg	Norinyl -1	£
* Equivalent to 35mcg ethinyl oestradiol			

Phasic pills

Oestrogen	Progestogen	Name	Cost
Ethinyl oestradiol 35mcg	First generation		
	Norethisterone 500mcg/1mg	Synphase	£
Ethinyl oestradiol 30/40mcg	Second generation		
	Levonorgestrel 50/75/125mcg	Logynon (ED)	£
		TriRegal	£

Oestradiol or estetrol pills

Oestrogen	4 th generation progestogen	Name	Cost
Oestradiol valerate 3/2/1mg (phasic)	Dienogest 2/3mg	Qlaira	££££££
Oestradiol hemihydrate 1.5mg	Nomegestrel 2.5mg	Zoely	£££££
Estetrol 14.2mg	Drospirenone 3mg	Drovelis	££££££

Other CHC preparations

Route	Oestrogen content	Third generation progestogen	Name	Cost
Transdermal patch	34mcg/d	Norelgestromin 200mcg/d	Evra	£££££
Vaginal ring	15mcg/d	Etonogestrel 120mcg/d	NuvaRing	££££££
			SyreniRing	££££££

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